



8047 - Extremely massive galaxies in the early universe? Confirming the nature of the most model-breaking object by hunting for stellar absorption features

Cycle: 4, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Bingjie Wang (PI)	Princeton University
Dr. Joel Leja (CoI)	The Pennsylvania State University
Dr. Weichen Wang (CoI) (ESA Member)	University of Milano-Bicocca
Ms. Josephine Frederika Willemijn Baggen (CoI)	Yale University
Prof. Karl Glazebrook (CoI)	Swinburne University of Technology
Prof. Pieter van Dokkum (CoI)	Yale University
Prof. Erica Nelson (CoI) (CoPI)	University of Colorado at Boulder
Prof. Ivo Labbe (CoI)	Swinburne University of Technology
Dr. Katherine Suess (CoI)	University of Colorado at Boulder
Dr. Sedona H. Price (CoI)	Space Telescope Science Institute
Dr. Anna G de Graaff (CoI) (ESA Member)	Max Planck Institute for Astronomy
Dr. Nikko J. Cleri (CoI)	The Pennsylvania State University
Dr. Tim Blake Miller (CoI)	Northwestern University
Dr. Rohan Naidu (CoI)	Massachusetts Institute of Technology
Dr. Rachel Bezanson (CoI)	University of Pittsburgh
Dr. Michaela Hirschmann (CoI) (ESA Member)	Ecole Polytechnique Federale de Lausanne
Prof. Michael Maseda (CoI)	University of Wisconsin - Madison
Dr. Jorrit Matthee (CoI) (ESA Member)	Institute of Science and Technology Austria
Prof. Jenny Emma Greene (CoI)	Princeton University
Dr. Harley Katz (CoI)	University of Chicago
Dr. David Setton (CoI)	Princeton University

<i>Name</i>	<i>Institution</i>
Mr. Andrea Weibel (CoI) (ESA Member)	University of Geneva, Department of Astronomy

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	2	q1-149-151-0	NIRSpec MultiObject Spectroscopy	(3) egs-all-v10-fin

ABSTRACT

The discovery of a population of red, compact objects at redshifts $z > 6$ is among the most surprising and significant findings from JWST. Initial stellar mass estimates from photometry suggest values as high as 10^{11} Msun, exceeding theoretical limits. Follow-up 48 min spectroscopic observations have only deepened the puzzle, revealing evidence that supports two drastically different interpretations: unexpectedly massive galaxies indicated by the observed Balmer breaks, which are signatures of old stars, or dust-reddened AGNs contributing to the red continuum, suggested by the presence of broad Balmer emission lines. The resulting uncertainty in the inferred stellar mass spans over 1 dex. To resolve this, we propose NIRSpec/G395M spectroscopy to measure velocity dispersions in stellar absorption lines in the brightest massive galaxy candidate at $z > 6$, RUBIES-49140. It also has the most extreme stellar mass of 10^{11} Msun if all the light is emitted by stars. A direct measurement of the dispersion in the most model-breaking object at the most discerning wavelengths is our best and only way forward. With a small investment of JWST time, we would be able to finally determine the origins of JWST's most enigmatic new population.

OBSERVING DESCRIPTION

We propose deep G395M/290LP spectroscopy of the bright massive galaxy candidate at $z=6.7$, RUBIES-49140. Our primary goal is to detect stellar absorption from the prominent Balmer break.

The G395M disperser provides the perfect compromise between sensitivity and spectral resolution. Our exposure time (14h) are tuned to ensure that we can distinguish between the normal (~ 200 km/s) and extreme (~ 1000 km/s) velocity dispersions.

We use the Multi-Shutter Array to increase the legacy value of the observations, and have other compact red objects and high-ionization line emitters as fillers.

We do not impose roll angle constraints, but request a minimum 20% background, given the deep, background-limited nature of the proposed observations.

Proposal 8047 - Targets - Extremely massive galaxies in the early universe? Confirming the nature of the most model-breaking object ...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(2)	49140	RA: 14 19 34.1395 (214.8922479d) Dec: +52 50 56.58 (52.84905d) Equinox: J2000		
	Comments: Description=[]				
Fixed Targets	(3)	egs-all-v10-fin	RA: 14 19 36.8588 (214.9035783d) Dec: +52 52 12.96 (52.87027d) Equinox: J2000		
	Comments: Description=[]				

Proposal 8047 - Observation 2 - Extremely massive galaxies in the early universe? Confirming the nature of the most model-breaking ...

Observation	Proposal 8047, Observation 2: q1-149-151-0											Thu Jan 22 22:00:32 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
Diagnostics	(q1-149-151-0 (Obs 2)) Warning (Form): Config q1-149-151-2 (#1) has 1 primary slit traces affected by failed open shutters.											
	(q1-149-151-0 (Obs 2)) Warning (Form): Config q4-134-145 (#2) has 2 master background shutters affected by failed open or closed shutters.											
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 2:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(3)	egs-all-v10-fin	RA: 14 19 36.8588 (214.9035783d) Dec: +52 52 12.96 (52.87027d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	Filter: CLEAR; Readout: NRSRAPIDD1; 8 sources in 3 quads; [Reduced Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD1	3	1	4	257.682		
	2	Filter: F140X; Readout: NRSRAPID; 8 sources in 4 quads; [Reduced Accuracy]	SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
	3	Filter: CLEAR; Readout: NRSRAPIDD2; 8 sources in 4 quads; [Reduced Accuracy]	SAME	CLEAR	Auto Acq MSA Config	NRSRAPIDD2	3	1	4	343.577		
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map		Spectral Overlap Threshold			
	MSATA	false	After Target ACQ	MSA Center	p1 (34 sources)		jwst-nirspec-hr-no-g235h		1.5			

Proposal 8047 - Observation 2 - Extremely massive galaxies in the early universe? Confirming the nature of the most model-breaking ...

Reference Stars	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	1	21452	214.937654	52.844561	24.05423164367675 8	1	56019	214.878275	52.881839	24.15316200256347 7	
	1	43921	214.867813	52.846230	24.18785667419433 6	1	431495	214.921086	52.843267	22.24745766693942	
	1	45663	214.909546	52.880952	23.14209938049316 4	1	455412	214.869655	52.857127	22.12113891950191 7	
	1	46133	214.920327	52.889683	24.10596847534179 7	1	990782	214.921253	52.845728	22.98573131469455 4	
	Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude	
	2	45760	214.913854	52.884145	21.53301811218261 7	2	458150	214.888654	52.876504	22.41312516559265 3	
	2	50386	214.931610	52.908696	21.29917907714843 8	2	990538	214.856741	52.891276	22.50047208038664	
	2	63831	214.860702	52.887044	21.91457176208496 2	2	990557	214.899331	52.934520	21.58382164636557	
	2	76019	214.883249	52.933389	21.52509689331054 7	2	990660	214.937419	52.920314	21.13340512408410 3	
Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude		
3	43921	214.867813	52.846230	24.18785667419433 6	3	75328	214.809456	52.878207	22.41029357910156 2		
3	61793	214.881538	52.897293	24.02708244323730 5	3	489073	214.843983	52.906421	23.92831650477048 2		
3	72631	214.856300	52.904969	24.48205184936523 4	3	991208	214.849950	52.853108	22.83214612429349		
3	74823	214.817985	52.882991	23.50416374206543 3	3	991673	214.866509	52.852722	22.64776934841417		
Confirmation	#	Confirmation Type		Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp		Conf. Total Integrations	Conf. Total Exposure Time		
	1	After Target Acq		NRSIRS2RAPID	4	1		1	72.944		
	2	After Target Acq		NRSIRS2RAPID	4	1		1	72.944		
	3	After Target Acq		NRSIRS2RAPID	4	1		1	72.944		
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G395H/F290LP)	q1-149-151-2	3 Shutter Slitlet	214.914333333333 333 Degrees 52.861247222222 225 Degrees	27.352955055278 283		0.15	3	12	18557.068
	2	2 (G395H/F290LP)	q4-134-145	3 Shutter Slitlet	214.89289541666 668 Degrees 52.906625 Degrees	27.335965710460 464		0.25	3	12	17681.735
	3	2 (G395H/F290LP)	8047.q3-307-148- 2	3 Shutter Slitlet	214.84755954166 667 Degrees 52.881704722222 224 Degrees	27.299823958422 23	-0.23	0.3	3	12	17681.735

Proposal 8047 - Observation 2 - Extremely massive galaxies in the early universe? Confirming the nature of the most model-breaking ...

Special Requirements	Group Visits within 53.0 Days Visits Same PA Background Limited. Background no more than 10th percentile above minimum MSA Scheduled Aperture PA 27.3444 to 27.3444 Degrees (V3 248.76982 to 248.76982)
----------------------	--